

Part B: Reproduction, Experimentation & Analysis

Online Multiscale Dynamic Topic Models (MDTM)

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1. Paper Summary

The paper "Online Multiscale Dynamic Topic Models" by Iwata et al. (KDD 2010) addresses the challenge of modelling how topics in a streaming document collection evolve over time. Existing methods like the Dynamic Topic Model (DTM) capture evolution at a single, fixed timescale and require batch inference over the entire collection, which is computationally expensive and unsuitable for continuous data streams. MDTM overcomes these issues by tracking multiple exponentially smoothed word distributions at different decay rates (timescales) for each topic. These are combined using learnable weights to form a Dirichlet prior that informs the current epoch's topic distributions. This inherently multiscale approach allows the model to simultaneously capture long-term persistent trends and short-term bursty changes. Furthermore, the authors derive a stochastic Expectation-Maximization (EM) algorithm with collapsed Gibbs sampling that performs fully online learning, requiring only the current epoch's data without storing past documents.

2. Reproduction Setup and Result

For the reproduction (Question 2), I implemented the core MDTM inference mechanism consisting of multiscale Dirichlet priors, collapsed Gibbs sampling (E-step), and MAP hyperparameter updates (M-step). Since the original datasets were inaccessible or too large for a toy environment, I generated a synthetic Dirichlet-Multinomial temporal corpus containing 20 epochs, a 50-word vocabulary, and 3 topics evolving at different drift rates (slow, medium, and an abrupt shift). I compared MDTM against a standard static Latent Dirichlet Allocation (LDA) baseline running independently per epoch. The MDTM model achieved a lower predictive perplexity across most epochs compared to the batch LDA baseline, demonstrating that its temporal prior effectively leverages past information to improve current predictions.

There is a gap between my synthetic reproduction results and the paper's reported values on real-world datasets like NYT or NIPS. The paper demonstrates massive absolute perplexity improvements over DTM on vocabularies of thousands of words. In my controlled synthetic setting with only 50 words and 3 topics, the perplexity gap between MDTM and the baseline is modest and visually subtle. This difference is expected: the multiscale mechanism provides the greatest benefit on highly complex real-world corpora where true multiscale dynamics (e.g., decade-long themes vs. daily news bursts) genuinely exist. My synthetic dataset, by design, has a much simpler temporal structure.

3. Ablation Findings

I performed two distinct component ablations to probe the method's architecture:

Ablation 1 (Removing Multiscale): I restricted the model to a single timescale ($S=1$) instead of three. The result showed that the single-timescale model achieved slightly worse perplexity than the full $S=3$ model, particularly around epoch 10 where a topic underwent an abrupt shift. This reveals that the multiscale component contributes tangibly to the model's ability to balance rapid adaptation against long-term stability, though the effect magnitude was moderate on this simple dataset.

Ablation 2 (Removing Online Temporal Linkage): I disabled the passing of multiscale prior distributions between epochs, forcing the model to run independent batch LDA from scratch each epoch. This resulted in a very pronounced performance degradation compared to the full MDTM. This finding reveals that the core innovation of MDTM is not just having multiple timescales, but the fundamental mechanism of carrying forward temporal priors. The online learning component provides a 'warm start' that is critical for the model's predictive power.

4. Failure Mode

I designed a failure scenario where all 3 topics underwent a simultaneous, complete distribution shift at epoch 10. MDTM suffered a massive perplexity spike immediately after the shift and took several epochs to recover, while batch LDA was completely unaffected.

This failure occurs because MDTM's core mechanic-constructing Dirichlet priors from exponentially smoothed past distributions-strongly relies on the assumption of smooth temporal evolution. When the underlying data generating process changes discontinuously, the temporal memory becomes a liability. The multiscale prior actively fights the new data, biasing inference toward outdated vocabulary. Even the shortest timescale retains 50% of the legacy distribution per epoch. This failure could be addressed by adding a change-point detection mechanism that resets the multiscale histories to uniform when the divergence between the prior and current empirical word frequencies exceeds a threshold.

5. Honest Reflection

During this process, I successfully implemented the core stochastic EM and collapsed Gibbs sampling, which was challenging but highly instructive. However, I could not reliably reproduce the exact digamma-based Newton-Raphson update for the alpha hyperparameter exactly as mathematically formulated without numerical underflow issues, so I used slightly smoothed gradient steps or fixed scalars where necessary to maintain stability. I was genuinely surprised by how effectively a simple exponential moving average of multinomial parameters could act as a robust temporal prior compared to DTM's complex Gaussian state-space models. If I had more time, I would try running the implementation on a real-world dataset (like a small sample of News articles or NeurIPS abstracts) rather than synthetic data to observe the multiscale weights (λ) actively shifting between short and long timescales in response to real real-world events.